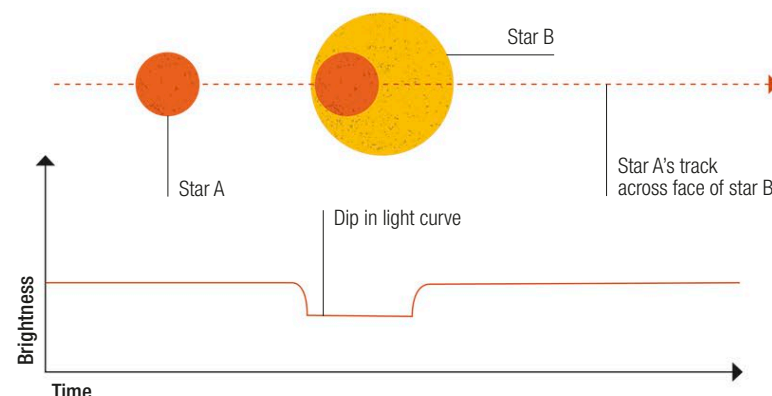


STAR SIZES

DESPITE APPEARING AS MERE PINPRICKS IN THE SKY, STARS DIFFER GREATLY IN SIZE, WITH MANY SO BIG THAT THEY DWARF OUR RELATIVELY SMALL SUN AND OTHERS SMALLER THAN SOME PLANETS IN OUR SOLAR SYSTEM.

The smallest stars are tiny, super-dense neutron stars that form after a giant star has collapsed. These stars are only 15 miles (25 km) in diameter. Most stars in our galaxy are dwarf stars, some of them with less than a thousandth of the Sun's volume. The largest stars, the super- and hypergiants, can be as much as 8 billion times greater in volume than the Sun. Stars are grouped into categories based on characteristics such as color, size, and brightness. A combination of color and brightness indicates a star's size. For example, a bright blue star is smaller than an equally bright red star, because a blue star is hotter than a red star and needs less surface area for it to be as bright as a cooler red star.



△ Measuring sizes

By examining the light curve during an eclipse in an eclipsing binary system (see p.43), it is possible to determine how long it takes for one star to pass the other, thereby making it possible to determine the diameters of the stars.

ORANGE GIANT Pollux

The orange coloration of Pollux indicates that it has a lower surface temperature than the Sun.

BLUE GIANT Bellatrix

Bellatrix is about 20 million years old and has a diameter six times that of the Sun.

YELLOW DWARF The Sun

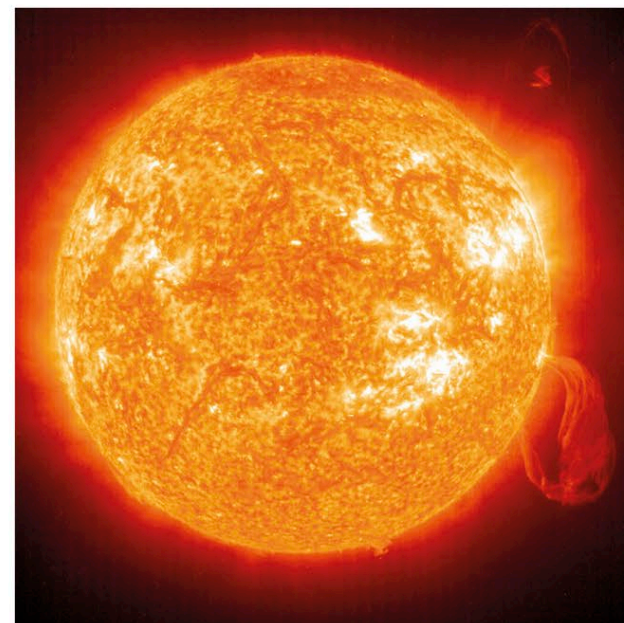
Stars in this category are all main-sequence stars and very similar in size to the Sun.

RED GIANT Aldebaran

Aldebaran is an irregular variable, meaning that its size changes from time to time as it tries to balance out forces of gravity and outward pressure.

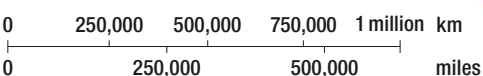
▷ Ordinary star

The Milky Way consists of at least 200 billion stars, of which 90 percent are in the stable stage (main sequence) of their life cycle. The Sun is a main-sequence star categorized as a yellow dwarf. It has a diameter of 864,000 miles (1.39 million km), but when it runs out of hydrogen it will swell into a red giant before losing its outer layers and finally becoming a white dwarf.



▷ Dwarf stars

Most stars are described as dwarf stars. This group of small, dim stars includes stars that are about the size of the Sun and many smaller red dwarfs and white dwarfs—tiny remnants of giant stars that have lost their outer layers. Brown dwarfs are bodies without enough mass to trigger nuclear fusion in their cores and are, in that sense, failed stars.



YELLOW DWARF The Sun

RED DWARF Proxima Centauri

Red dwarfs are the most numerous star type in our galaxy and will eventually also become white dwarfs.

BROWN DWARF EROS-MP J0032-4405

Not actually stars, most brown dwarfs are about the same size as the planet Jupiter in our Solar System.

WHITE DWARF Sirius B

Sirius B is roughly the same size as Earth, but its mass is nearly equal to that of the Sun.